

HDB Resale Price Prediction

ICT6001 Week 01 Capstone — Group 4

Objective

Predict HDB resale prices from transaction-level resale flat data, then explain model errors and business usage limits.

82,801

Raw rows

11

Columns

4

Final model features

Team member	Role / responsibility
Mirza	Data Understanding + Data Cleaning
Divya	Modeling (Linear Regression)
Kat Long	Modeling (Random Forest)
David	Cross Valuation + Final Test + Failure Analysis

Submission links

Google Colab:
<https://colab.research.google.com/drive/1wfEtACl2304TE8qM8lOi76gww--mmH2-?usp=sharing>
YouTube: <https://youtu.be/DwTu2r8cGH4>

Executive Summary

Random Forest is the champion model; use it as a decision-support estimate, not a definitive valuation.

\$57,403

Final RF MAE

\$83,186

Final RF RMSE

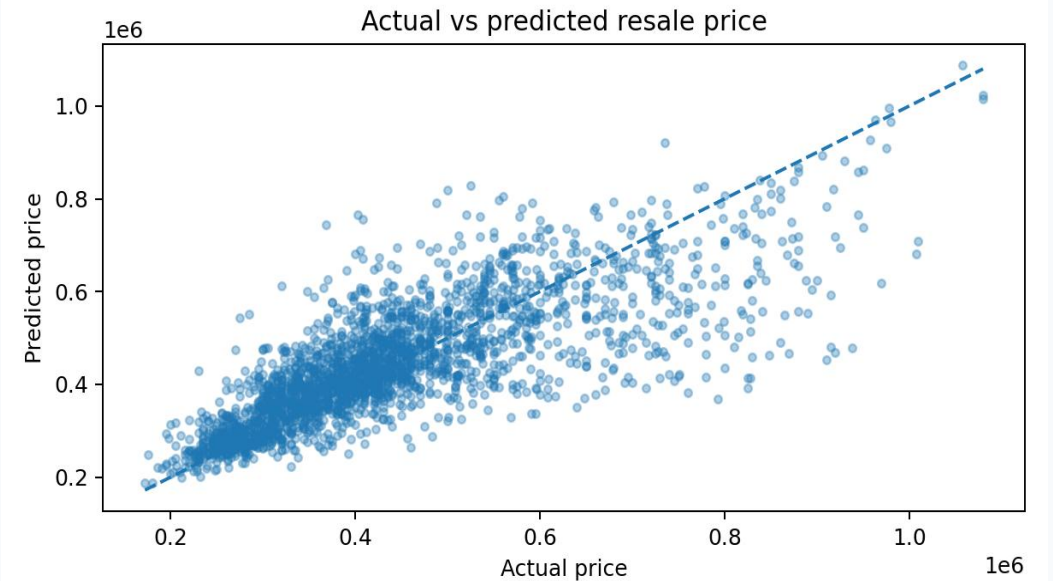
0.708

Final RF R^2

12.7%

Final RF MAPE

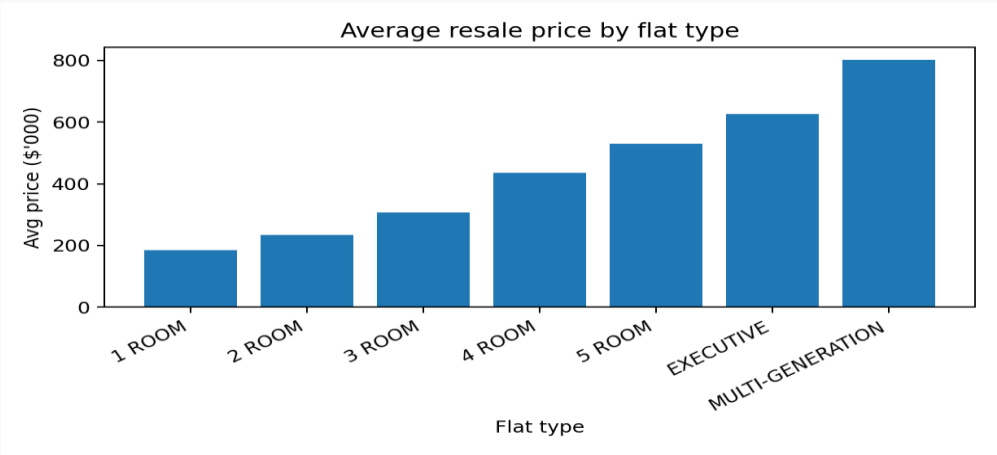
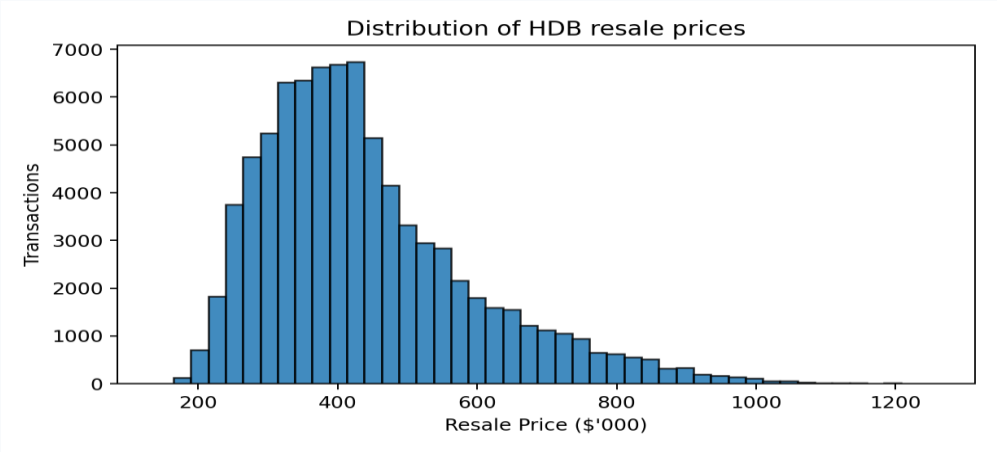
- Regression task: predict resale_price from cleaned HDB resale transaction data.
- Cleaning removes missing rows and duplicate rows before modelling.
- Random Forest beats Linear Regression on MAE, RMSE, R^2 and MAPE.
- Main errors occur for towns and flat types not directly encoded in the final feature set.



Data Engineering

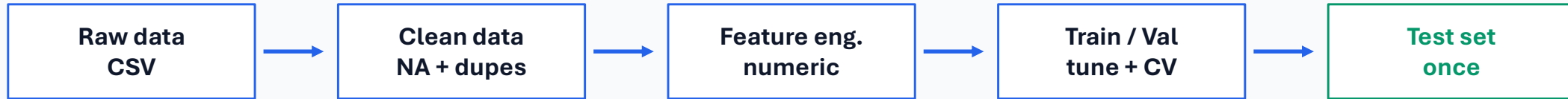
Data understanding, data cleaning and feature creation before modelling.

Step	Action	Why it matters
Raw input	82,801 rows × 11 columns	Starting resale transaction dataset
Quality checks	Check missing values + duplicates	Avoid dirty rows affecting training
Clean data	drop_duplicates() + dropna()	Removes 211 duplicates and 47 missing values
Date features	month → sale_year / sale_month	Captures timing of resale transaction
Storey feature	storey_range → storey_mid	Turns floor range into numeric value
Lease feature	remaining_lease → months	Makes remaining lease model-readable



Pipeline Architecture

How we reduced data leakage risk in the modelling workflow.



- Train/validation/test split is created before model comparison.
- Validation set is used for model selection and hyperparameter tuning.
- Cross-validation is run on training data only.
- Final test set is used once after the champion model is chosen.
- Linear Regression CV uses Pipeline(StandardScaler + LinearRegression).

Model Selection

Linear Regression baseline. Random Forest comparison.

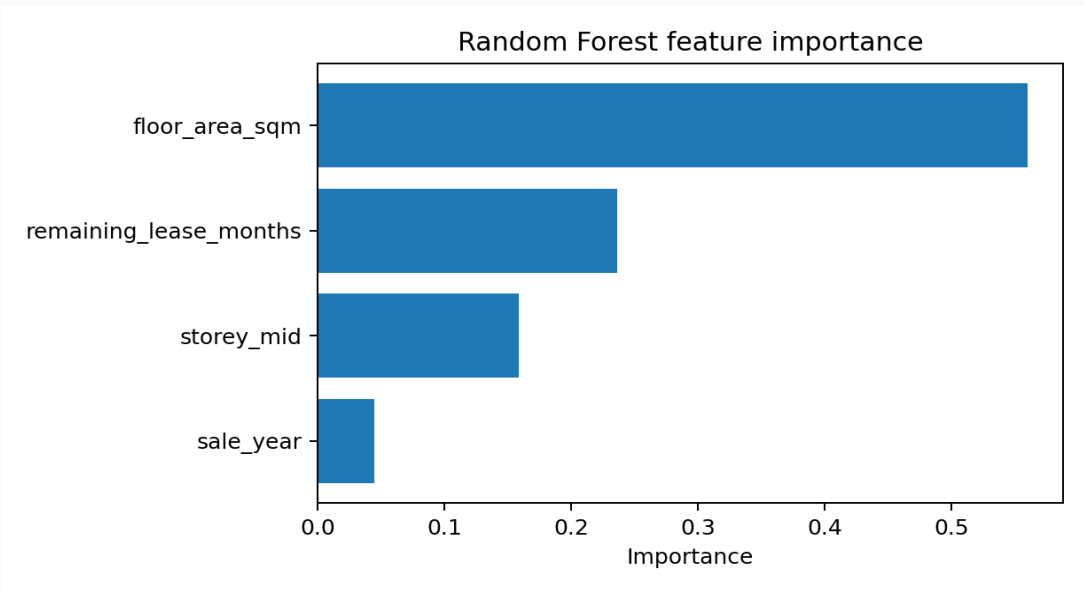
Model	MAE ↓	RMSE ↓	R ² ↑	MAPE ↓
Linear Regression	\$77,310	\$103,996	0.543	17.1%
Random Forest	\$57,403	\$83,186	0.708	12.7%

Selection logic

- Linear Regression gives a simple baseline and is easy to explain.
- Random Forest better captures non-linear relationships in resale prices.
- Champion model selected because it reduces dollar error and improves R².
- Only models compared in the notebook are reported; no hidden test-set tuning.

Justification of the Champion Model

Random Forest Regressor gives stronger price prediction than the baseline.



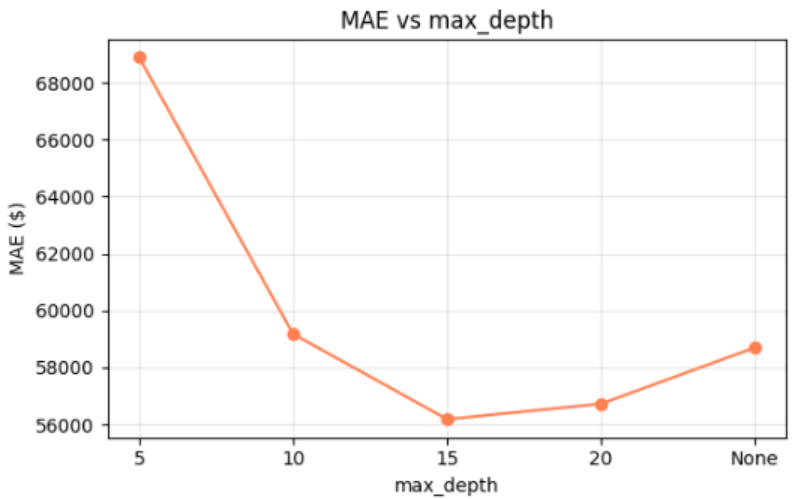
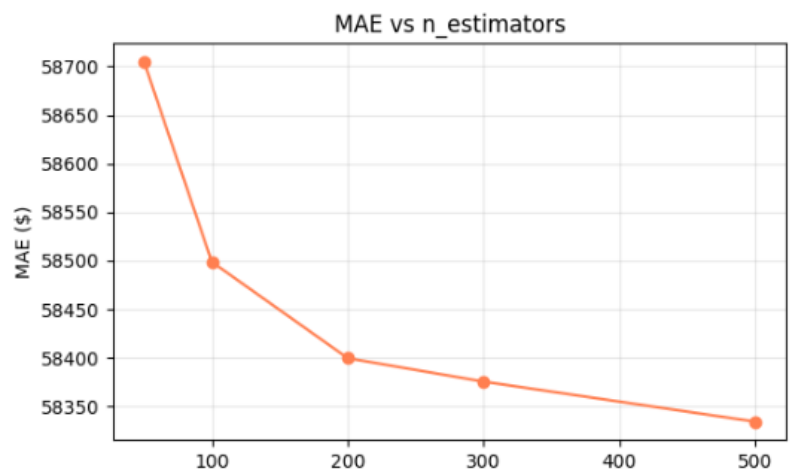
- Champion: RandomForestRegressor.
- Final feature inputs: floor_area_sqm, storey_mid, sale_year, remaining_lease_months.
- Random Forest can model interactions such as area × lease × storey.
- Limitation: current features do not include town or flat_type, so premium-location effects are missed.

Why Linear Regression underperforms	Why Random Forest is better here
Assumes mostly straight-line effects	Captures curved/non-linear effects
Misses interaction effects	Learns feature interactions
Higher error in comparison table	Lower MAE/RMSE and higher R ²

Controlled Ablation & Tuning Log

Tuning changed one main idea at a time: baseline, scaling, trees, depth.

Experiment	Hypothesis	Result / decision
Baseline Linear Regression	Simple numeric features can predict price	Kept as baseline model
Standardised Linear Regression	Scaling may help linear model	Similar performance; remove duplicate block
RF n_estimators	More trees can reduce variance	Tuned across 50, 100, 200, 300, 500
RF max_depth	Depth controls underfit/overfit	Tuned across 5, 10, 15, 20, None
Failure feature check	town / flat_type explain failures	Use as proposed improvement



Final Model Stability Metrics

Current comparison shows Random Forest is stronger; add std output in Colab for exact mean ± std.

Model	CV MAE	CV RMSE	CV R ²	CV MAPE
Linear Regression	\$77310	\$103996	0.543	0.171
Random Forest	\$57403	\$83186	0.708	0.127

Final reported evaluation

\$56,431

RF MAE

\$82,398

RF RMSE

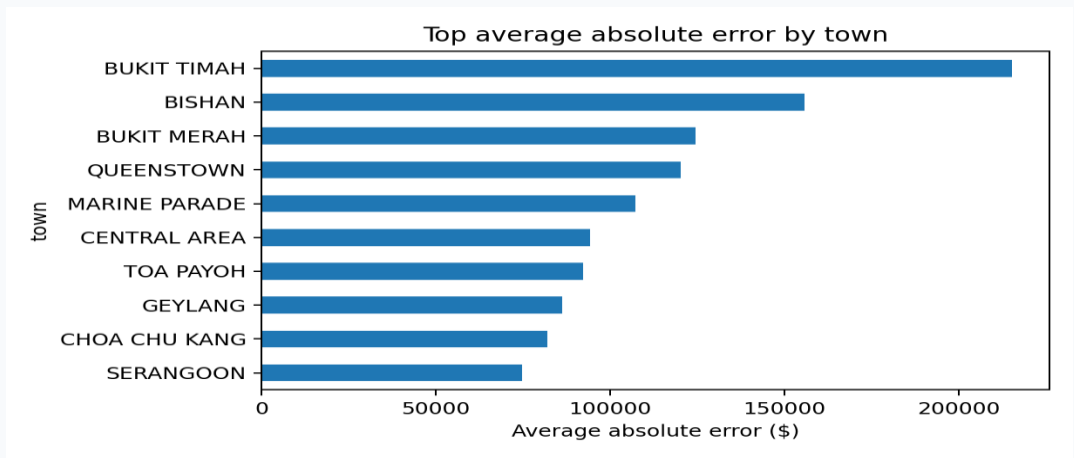
0.717

RF R²

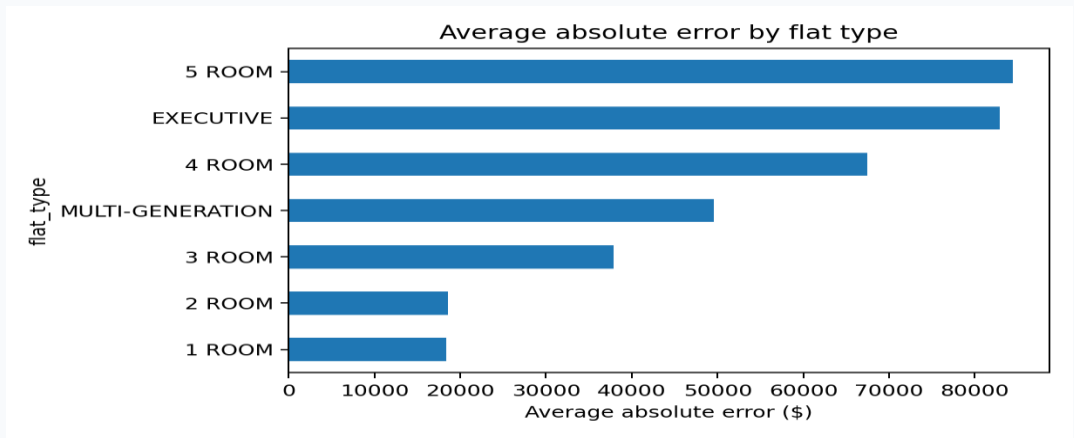
12.45%

RF MAPE

Mechanical Failure Analysis



Town	Type	Actual	Predicted	Abs. error
QUEENSTOWN	5 ROOM	\$938,000	\$478,782	\$459,218
GEYLANG	5 ROOM	\$910,000	\$452,542	\$457,458
CENTRAL AREA	4 ROOM	\$920,000	\$468,155	\$451,845
GEYLANG	5 ROOM	\$915,000	\$480,582	\$434,418
TOA PAYOH	5 ROOM	\$825,000	\$392,767	\$432,233



- Worst rows are mostly underpredictions in premium or high-demand towns.
- Larger flat types show higher average absolute error.
- Reason: town and flat_type are used in analysis but not used as model input features.
- This supports the next improvement: encode categorical location and flat-type features.

Proposed Fixes for Observed Failures

Fixes are directly linked to the error patterns, not generic “try harder” improvements.

Failure observed	Likely cause	Proposed technical fix
Premium towns underpredicted	town not in final feature set	OneHotEncode town or group towns by region/maturity
Larger flats have higher error	flat_type not used in model	Encode flat_type and/or flat_model
Time market trend limited	sale_year is coarse	Use sale_month_index for month-level trend
Outlier rows hard to predict	unusual flats / premium demand	Flag high-error cases for manual review
Repeated code blocks	manual metrics and plots	Use reusable evaluate_model() and plot function

Highest-priority next iteration

Build a leakage-safe ColumnTransformer + Pipeline: numeric features pass through; categorical features such as town, flat_type and flat_model are OneHotEncoded. This addresses the largest observed errors while preserving correct train/test separation.

Business Decision Reasoning

Different users suffer different costs from overprediction and underprediction.

Prediction error	Who is harmed?	Impact	Recommended display rule
Overprediction	Buyer	Buyer may overpay above fair market value	Show conservative buyer estimate
Underprediction	Seller	Seller may price too low and lose potential value	Show price range, not one fixed price
High uncertainty	Both sides	Unusual flats reduce trust in prediction	Flag for manual review

Buyer-facing safety margin

Displayed estimate = predicted price – one test MAE $\approx$ predicted price ~ \$57K.

Use case boundary

Use this as an early decision-support tool. Do not present it as an official HDB valuation, bank valuation, or guaranteed market price.

Individual Contributions

Mandatory slide: individual tasks, not generic group-level work.

Member	Individual task completed	Evidence in report
Mirza	Data understanding, data cleaning, feature engineering checks	Slides 1, 2, 3, 4
Divya	Linear Regression baseline and comparison metrics	Slides 5, 6, 7, 8
Kat Long	Random Forest model, hyperparameter tuning and champion selection	Slides 5, 6, 7, 8
David	Evaluation metrics, row-level error inspection and business reasoning	Slides 9, 10, 11, 12